

Name: Voot Tangkaratt
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Current address: Tokyo, Japan

I am a robot learning researcher and engineer on **imitation and reinforcement learning for robotics**.
My career goal is to make physical AI and robots that are useful to everybody.
I am from Thailand and have been living in Japan since 2012.

Technical interests

- Real-world physical AI development with imitation and reinforcement learning.
- Robot learning algorithms for multi-tasks and fast adaptation to unseen tasks.

Work history

- **Senior Research Scientist** at Sony Research/Sony AI, Japan, 2022 - 2026
 - **Responsibility:** Research and development of **robot learning pipeline, models, and algorithms** for fine manipulation in gastronomy robotics.
- **Postdoc Researcher** at RIKEN AIP, Japan, 2017 – 2021
 - **Responsibility:** Research and development of **robust imitation and reinforcement learning models and algorithms** under real-world data constraints.

Education

- **PhD in Computer Science**, The University of Tokyo, Japan, 2014-2017.
- **Masters in Computer Science**, Tokyo Institute of Technology, Japan, 2012-2014.
- **Bachelors in Computer Engineering**, Chulalongkorn University, Thailand, 2008-2011.
- **High school:** Suankularb Wittayalai School, Thailand, 2002-2008.

Technical skills

- **Research and development:**
 - Expertised in **research and development of imitation** (behavioral cloning, GAIL, VLA) and **reinforcement learning** (e.g., actor-critic, model-based RL, offline RL) models and algorithms.
 - Experience in **training and deploying learning models on real-world robots**.
 - Experience with **reward design and reward learning** (e.g., inverse reinforcement learning and preference reward learning).
 - Experience in **research and development of robust ML** (e.g., robust optimization and uncertainty estimation).
 - Experience in **implementing deep learning models and architectures**.
 - Experience in **implementing a robot learning pipeline** on real-world robots with Python
 - Experience in **collecting real-world robot data of proprioceptor, vision, and tactile sensors** using kinesthetic teaching, tele-operation, and motion capture systems.
 - Familiarity with **using GPU clusters for model training**.
 - Familiarity with **using position and impedance controllers** for controlling robot manipulators.

- **Programming:**
 - Experience in **Python** programming.
 - Experience in using **deep learning libraries** (Pytorch, Jax, Keras).
 - Experience with **Robot Operating System (ROS)** 1 and 2.
 - Experience with **robot simulation libraries** (Gymnasium, Mujoco, and Pybullet).
 - Familiarity with software development tools including AWS, Git, and Docker containers.
- **Language:**
 - Academic and business-level English, Intermediate-level Japanese, and Native Thai.

Non-technical skills

- Experience in managing roadmaps and development of a small group of engineers and researchers.
- Experience with collaborating with external organizations for research, publications, and patents.

Selected publications (For a full publication list, please visit [google scholar](#) or [my github page](#))

- Robust Imitation Learning from Noisy Demonstrations
 - **Voot Tangkaratt**, Nontawat Charoenphakdee, Masashi Sugiyama
 - In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021.
- Variational Imitation Learning with Diverse-quality Demonstrations
 - **Voot Tangkaratt**, Bo Han, Mohammad Emtiyaz Khan, Masashi Sugiyama
 - In *International Conference on Machine Learning (ICML)*, 2020.
- Simultaneous Planning for Item Picking and Placing by Deep Reinforcement Learning
 - Tatsuya Tanaka, Toshimitsu Kaneko, Masahiro Sekine, **Voot Tangkaratt**, Masashi Sugiyama
 - In *International Conference on Intelligent Robots and Systems (IROS)*, 2020.
- Imitation Learning from Imperfect Demonstration
 - Yueh-Hua Wu, Nontawat Charoenphakdee, Han Bao, **Voot Tangkaratt**, Masashi Sugiyama
 - In *International Conference on Machine Learning (ICML)*, 2019.
- TD-regularized actor-critic methods
 - Simone Parisi, **Voot Tangkaratt**, Jan Peters, Mohammad Emtiyaz Khan
 - In *Machine Learning journal (MLJ)*, 2019.
- Fast and Scalable Bayesian Deep Learning by Weight-perturbation in Adam
 - Mohammad Khan, Didrik Nielsen, **Voot Tangkaratt**, Wu Lin, Yarin Gal, Akash Srivastava
 - In *International Conference on Machine Learning (ICML)*, 2018.
- Guide actor-critic for continuous control
 - **Voot Tangkaratt**, Abbas Abdolmaleki, Masashi Sugiyama
 - In *International Conference on Learning Representations (ICLR)*, 2018.